

[DC] Innovative XR Approaches for Behavioral and Cognitive Therapy

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ABSTRACT

The increasing deployment of immersive technologies is opening up opportunities in the field of behavioral and cognitive therapies. In particular, the use of eXtended Reality (XR) technologies allows the development of therapies that require moving users to a remote location without the need to go to the therapy site. This is especially useful when the therapy is aimed at phobia treatment or attention training skills. Specifically, this PhD aims to research innovative therapy sessions by incorporating XR-based technologies, assessing their effectiveness in treating conditions like phobias, enhancing cognitive abilities, and improving the accessibility of therapy for individuals with psychosocial difficulties, such as intellectual disabilities.

Index Terms: eXtended Reality, Accessibility, Intellectual disability

1 INTRODUCTION

The development of eXtended Reality (XR) technologies has created new possibilities in therapy. Specifically, XR has proven to be a potential tool in improving behavioral and cognitive therapies, providing innovative treatments for conditions such as phobias and anxiety disorders [6]. These immersive technologies allow users to interact with virtual environments, creating a secure and controlled setting for therapeutic activities. It's important to assess users effectively in XR-based therapies to make sure these technologies meet individual needs and lead to positive therapeutic results.

One of the key advantages of XR-based therapies is the ability to simulate real-world situations without needing patients to physically be present at therapy locations. This feature is especially beneficial for gradual exposure to fear-inducing stimuli in the therapeutic process [7]. Furthermore, XR tools are becoming increasingly relevant for individuals with intellectual disabilities, offering customized and adapted therapeutic approaches that can address specific cognitive and behavioral challenges.

The aim of this PhD is to explore the use of XR technologies in therapeutic contexts, focusing on their applications in behavioral and cognitive therapies. As a starting point, it examines the effectiveness of XR-based systematic desensitization for phobia treatment and immersive cognitive training on attention in individuals with intellectual disabilities. It also explores the accessibility and usability of XR systems, guaranteeing that they are inclusive to the diverse needs of users. Through this research, we aim to demonstrate how XR technologies can advance therapeutic practices.

2 CONTEXT

Immersive technologies, particularly VR, have gained significant attention for their application in exposure therapy. This approach builds on the foundations of traditional exposure therapy by immersing patients in virtual environments that closely resemble real-world situations [2]. Research highlights the positive outcomes of blending conventional psychological practices with VR to address

fears and phobias. Garcia-Palacios et al. [7] reported that many individuals prefer therapy involving immersive technologies over traditional methods. This preference can be attributed to the safer, less stigmatizing, and often more cost-effective nature of XR-based exposure to phobic stimuli compared to standard approaches.

Research [1] has shown VR's potential in treating acrophobia (fear of heights). Since then, VR has been found effective to address a range of anxiety disorders, including aviophobia (fear of flying) [10], social anxiety disorder [6], and other phobias. For example, a study on social phobia demonstrated that combining XR with cognitive behavioral therapy (CBT) offers promising results as an innovative treatment strategy [9].

Research has also highlighted the potential of VR in the treatment of various conditions, including functional neurologic disorder (FND) [3]. FND is a complex condition that causes neurological symptoms without an apparent structural problem in the nervous system. For instance, Bullock et al. [4] conducted a study using VR-based mirror visual feedback therapy. In this study, participants underwent VR exposure therapy based on self-identified FND triggers.

Lastly, user experience in XR technologies has traditionally been evaluated through questionnaires to assess factors like presence [11], interaction and emotional states such as stress or anxiety. However, these question-based methods have limitations due to their complexity and subjective nature. Therefore, there is a need for more inclusive assessment methods [8] that are simplified to increase understanding for all users.

3 OBJECTIVES OF THE RESEARCH

The goal of this research is to improve therapy with XR technologies, making them accessible to individuals with psychosocial difficulties. This includes simplified interfaces, adaptive systems, and customized scenarios to meet users' needs. Additionally, this research aims to develop inclusive evaluation techniques to assess user experience and the effectiveness of XR-based treatments. The research focuses on XR applications in the following therapy domains:

1. **Behavioral Therapy:** XR can simulate real-world environments in a safe, controlled space, allowing individuals to confront and manage behavioral challenges. This includes treatments for phobias, such as bathmophobia (fear of stairs).
2. **Cognitive Therapy:** XR tools can create cognitive training tasks, particularly for improving attention. By offering interactive environments, XR provides innovative ways for users to practice these skills.
3. **Exploration of Other Therapies:** This research also explores XR's potential in therapies for other conditions, such as neurological dysfunctions and developmental disorders.

The research questions resulted from these objectives are:

1. How effective is XR-based therapy in treating phobias compared to traditional methods?
2. How effective is XR-based cognitive training compared to traditional methods?

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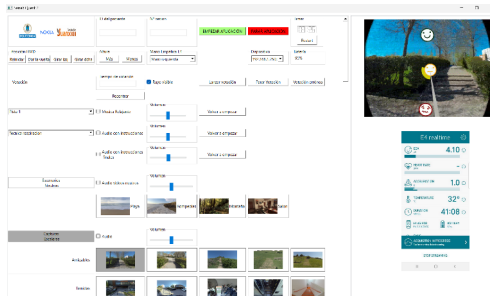


Figure 1: Therapist interface with control and visualization areas

3. Are XR applications sufficiently inclusive to ensure accessibility for diverse users, including individuals with psychosocial difficulties?
4. Are existing methods, including questionnaires, valid for evaluating user experience and the effectiveness of XR-based treatments?

4 CURRENT STATUS AND FUTURE WORK

The ongoing research focuses on developing a system to support behavioral and cognitive therapy sessions and on evaluating user experience and therapy effectiveness through inclusive methods. The system consists of two components: a therapist interface on a computer and a user application on a head-mounted display (HMD). The therapist interface manages the session, with a control area and a visualization area to mirror the user's HMD view to guide the user's experience. Evaluation includes designing questionnaires and analyzing feedback.

The developed behavioral therapy application addresses bathmophobia, based on research in [5]. Using systematic desensitization, users start with relaxing virtual scenarios and are gradually exposed to phobic stimuli through 360-degree videos and VR scenes. An adapted voting system tracks anxiety levels, where users select one of three faces (green, yellow, or red) with head movements. This allows monitoring of anxiety progression during sessions. Figure 1 shows the therapist's interface with the voting system.

The study was conducted at an Occupational Center for Training, Opportunities, and Labor Integration, which supports individuals with intellectual disabilities. Ten volunteers participated in a therapy program that lasted six months and was implemented and evaluated by the center's therapists. For evaluation, adapted questionnaires were conducted to the users. At the end of the program, this study showed that the use of systematic desensitization technique mixed with XR helped the users overcome their fear¹.

For cognitive therapy, a second application is being developed. This focuses on training attention skills for individuals with intellectual disabilities. Two primary scenarios have been created: a cafeteria and a supermarket. Each scenario presents task-based activities designed to train on attention. In the cafeteria scenario, tasks include collecting dirty dishes and serving customer orders. In the supermarket scenario, users are tasked with restocking products and processing customer orders.

For future work, the data from the behavioral therapy experiments will be analyzed. A study will also be conducted at an Occupational Center to evaluate the system and its integration with the therapeutic techniques used in cognitive therapy. The system is adaptable to treat a variety of phobias beyond bathmophobia, and new scenarios can be designed for other conditions. Additionally,

¹<https://www.youtube.com/watch?v=qs-FrPsmH8>

the framework can be extended to explore the potential of XR technologies in diagnosing and treating conditions like Functional Neurological Disorder (FND), which is under-researched, and could help with its detection and treatment.

5 CONCLUSION

This research aims to explore the potential of XR technologies in advancing behavioral and cognitive therapies, focusing on treating conditions like phobias and attention deficits, while ensuring accessibility for individuals with psychosocial difficulties.

Next steps include analyzing data and testing the cognitive therapy system to assess the effectiveness and scalability of XR therapies. The goal is to make XR technologies more accessible and useful in a variety of therapeutic contexts, including neurological disorders like FND.

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